Multiple-Test Admissions

A joint analysis and corrected version of two Grady Towers letters

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Grady Towers's first letter gives incorrect overlap probabilities for correlated IQ tests; his follow-up explicitly retracts that calculation and introduces a substantially better numerical method. This report assesses the two together. At correlation 0.7, a bivariate-normal model with two top-2% cutoffs gives 3.3724% eligible through either test and 31.3814% repeat qualification among passers of one specified test. For top-2% and top-0.1% cutoffs on different tests, 65.8589% of higher-threshold passers also pass the lower-threshold test. The follow-up's remaining numerical differences arise chiefly from grid boundaries, rounded cutoffs, and inconsistent marginal denominators. Neither letter establishes actual membership proportions or a latent-ability interpretation. A corrected combined article and reproducible calculations are provided.

1 Joint assessment of the two sources

1.1 Read the second letter as a retraction

The sources are *On using multiple tests for high IQ society admissions* (Towers) and its *Followup* (Towers, 2000), both hosted by Darryl Miyaguchi. The first is undated in the supplied page. The second identifies its underlying letter as dated 12 January 2000 and says that it was published posthumously with permission. The supplied Wayback captures are dated 9 March 2026 and 5 May 2022, respectively; these are archive dates, not the letters' publication dates.

The follow-up expressly withdraws the earlier calculation. It should not be presented as independent support for the first letter's dramatic conclusion. Towers's useful qualitative point survives: accepting a pass on either of two imperfectly overlapping tests generally enlarges the eligible group relative to either test alone. The numerical examples and the interpretation of eligibility need correction.

This report examines the mathematical scenarios in those historical letters. It does not establish the admission policies, test correlations, or membership composition of any named society, either historically or today. The incomplete 1960 reference in the first letter can be identified as Zelen and Severo (1960); the calculations here were performed independently.

1.2 The first letter's principal numerical claims fail

Table 1 compares the first letter's principal figures with recalculations under its intended Gaussian scenario. In the mixed case, A means passing the top-2% cutoff on test X, and B means passing the top-0.1% cutoff on a different test Y. All corrected entries assume population score correlation $\rho = 0.7$.

Table 1: Selected claims in the original letter and corrected model results. All displayed probabilities are percentages, with the denominator specified.

Quantity	**Original**	**Corrected**
Both top-2% tests: population	about 0.2	0.627627
Either top-2% test: population	3.83	3.372373
Second top-2% pass, given first pass	10.27	31.381370
Mixed case: either pass, population	2.11	2.034141
Mixed case: both pass, population	about 0.014	0.065859
Mixed case: higher cutoff given lower, $P(B \mid A)$	0.69	3.292944
Mixed case: lower cutoff given higher, $P(A \mid B)$	13.48	65.858880

One error is visible without a distributional model: the mixed-case union cannot be 2.11% when its marginal probabilities sum to only 2.10%. In general, $P(A \cup B) \leq P(A) + P(B)$. The corrected mixed-case failure rate on the lower-threshold test among higher-threshold passers is 34.1411%, rather than the original's approximately 86%.

For the equal top-2% case, passing X alone has probability 1.3724%, and passing Y alone has the same probability. Passing *exactly one* therefore has probability 2.7447%. These are different quantities; wording such as "one test but not the other" should specify which is intended.

1.3 What the follow-up fixes, and what remains

The follow-up replaces an approximation involving the phi coefficient with numerical summation of a bivariate-normal density. That is a substantial correction. Reimplementing its supplied BASIC grid closely reproduces its published joint probabilities. The remaining issues are:

- The equal-weight half-IQ-point grid samples the cutoff itself, giving approximately the boundary of a midpoint cell a quarter point below that cutoff. It does not integrate exactly from the stated continuous cutoff.
- The integer IQ cutoffs 133, 150, and 164 do not have exactly the nominal tail probabilities used as denominators in the follow-up.
- A conditional proportion among passers of one specified test is not automatically a proportion among society members, or even among all people eligible through either of two tests.

Section A separates those effects. The absent scale and cell-area factors in the density ratio cancel; they are not fatal normalization errors. The image of the follow-up's double integral was also inspected, in addition to its HTML text and BASIC listing.

The follow-up recounts a top-5% example attributed to page 306 of Cattell's *The Scientific Analysis of Personality*. Under the model used here, two such tests at $\rho = 0.7$ have joint probability 1.95993%, rather than 1%. This checks Towers's recollection as written. Cattell's cited page was not independently inspected, so no error is attributed to Cattell himself.

2 Corrected combined article: admissions through correlated tests

An admission rule and a test percentile answer different questions. A test cutoff may be exceeded by 2% of a reference population. A rule accepting either of two such cutoffs concerns the union of two sets of people, whose size depends on their overlap. Requiring both cutoffs concerns their intersection. Neither rule, by itself, identifies the people with the highest values of a perfectly measured underlying ability.

2.1 Specify a joint model and distinguish the correlations

Let X, Y denote standardized scores in the same reference population, with a bivariate-normal distribution:

$$\begin{pmatrix} X \\ Y \end{pmatrix} \sim N_2\left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}\right], \qquad -1 < \rho < 1. \tag{1}$$

Here ρ is the Pearson correlation of the continuous scores. It is not the reliability of either test and is not generally the correlation of the binary pass/fail indicators. Normal marginal distributions and a correlation alone do not establish equation (1); the joint distribution is an additional assumption, especially consequential in the tails.

For desired marginal pass probabilities p_A, p_B, use

$$a = \Phi^{-1}(1 - p_A), \qquad b = \Phi^{-1}(1 - p_B), \qquad A = \{X > a\}, \quad B = \{Y > b\}. \tag{2}$$

The joint pass probability follows by conditioning on X:

$$\boxed{J = P(A \cap B) = \int_a^\infty \varphi(x)\, \overline{\Phi}\left(\frac{b - \rho x}{\sqrt{1 - \rho^2}}\right) dx.} \tag{3}$$

Here φ is the standard-normal density and $\overline{\Phi} = 1 - \Phi$ its upper-tail function. The remaining quantities are

$$P(A \cup B) = p_A + p_B - J, \tag{4}$$

$$P(B \mid A) = J/p_A, \qquad P(A \mid B) = J/p_B. \tag{5}$$

The denominators determine the question being answered.

For binary indicators $1_A, 1_B$, the phi coefficient is

$$\phi_{\text{binary}} = \frac{J - p_A p_B}{\sqrt{p_A(1 - p_A)p_B(1 - p_B)}}. \tag{6}$$

This differs from the correlation of the underlying continuous Gaussian variables (Ekström, 2011). For equal top-2% thresholds at $\rho = 0.7$, $\phi_{\text{binary}} \approx 0.29981$, rather than 0.7. The follow-up attributes its earlier error to a phi/product-moment approximation. Without the first algorithm, its exact computational mistake cannot be reconstructed; substituting one correlation for the other is nevertheless unjustified.

2.2 Two tests with the same nominal cutoff

For two top-2% cutoffs and $\rho = 0.7$, equation (3) gives

$$J = 0.0062762740205, \qquad P(A \cup B) = 0.0337237259795. \tag{7}$$

Thus 0.627627% of the population passes both tests, and 3.372373% passes at least one. Among people who pass a specified first test,

$$P(B \mid A) = \frac{0.0062762740205}{0.02} \approx 31.3814\%. \tag{8}$$

This statement concerns repeat qualification, not test reliability and not an unconditional probability of joining a society.

If eligibility instead means passing *either* of the two tests, the fraction of that eligible union passing both is

$$P(A \cap B \mid A \cup B) = \frac{J}{2p - J} \approx 18.6109\%, \qquad p = 0.02. \tag{9}$$

The 31.3814% and 18.6109% figures are both correct under the model; they use different denominators. Actual members may be a further selected subset of either group.

The same calculation at rarer equal cutoffs yields table 2. The cutoffs are exact nominal percentiles rather than rounded IQ points. IQ-style values in the table use mean 100 and standard deviation 16; they are scale conversions.

Table 2: Equal nominal cutoffs at $\rho = 0.7$: population overlap and repeat qualification among passers of one specified test.

Marginal tail on each test	**z cutoff**	**SD-16 cutoff**	**Both: population %**	**Second pass given first: %**
2%	2.053749	132.85998	0.627627	31.381370
0.1%	3.090232	149.44372	0.0159589	15.958919
1/33,000	4.010439	164.16702	0.000231250	7.631243

2.3 A lower cutoff on one test and a higher cutoff on another

Let $p_A = 0.02$ and $p_B = 0.001$, still with $\rho = 0.7$. The joint probability is

$$J = 0.0006585888039. \tag{10}$$

Consequently,

$$P(B \mid A) = 3.292944\%, \qquad P(A \mid B) = 65.858880\%, \tag{11}$$

$$P(A^c \mid B) = 34.141120\%, \qquad P(A \cup B) = 2.034141\%. \tag{12}$$

The asymmetry is expected: the higher-threshold passers are a much smaller group. These two events concern *different tests*. If both cutoffs were applied to the same score, the higher-cutoff event would be contained in the lower-cutoff event and the corresponding conditional probability would be 100%.

The result is sensitive to the assumed correlation. For these same mixed cutoffs, $P(A \mid B)$ is approximately 33.59% at $\rho = 0.5$, 65.86% at 0.7, and 97.99% at 0.9. A nominal 0.7 is an illustrative parameter, not a

universal empirical property of IQ tests. Population, norms, age, constructs, administration, and selection all matter.

These conditional model results cannot determine whether someone who passed a particular magazine test could join a particular society. That requires evidence about the actual instruments and the society's accepted routes. Nor does the model establish the earlier claim about the prior memberships of most high-scoring society members.

2.4 Eligibility size, rarity equivalents, and calibrated rules

The population fraction admitted by a rule can be expressed as a normal rarity equivalent. Under the equal top-2% example, the OR and AND fractions map to approximately 129.26 and 139.94, respectively, on an SD-16 scale:

$$I_{\text{rarity}} = 100 + 16\Phi^{-1}(1 - p_{\text{eligible}}). \tag{13}$$

This only matches a selected fraction. It does not show that the rules select the same people as a single test at that cutoff, establish "super-high reliability," or identify a particular level of latent intelligence. The first letter's corresponding AND rarity equivalent of about 146 follows from its erroneous overlap probability.

Multiple tests do not make a 2% overall eligibility rule impossible. Within the stated model, one can choose the common cutoff a_* to solve

$$2\overline{\Phi}(a_*) - J(a_*, a_*; 0.7) = 0.02. \tag{14}$$

The solution is approximately $a_* = 2.270055$, or 136.32088 on an SD-16 scale. Each test then has marginal pass rate approximately 1.16021%, while the two-test OR rule has overall rate 2%. This is an example of calibrating a *rule*; using more than one test is not inherently incompatible with a specified eligible fraction.

2.5 Interpret the population model and the testing process

A useful report separates three questions: who exceeds a cutoff on one instrument; who is eligible under a specified admission procedure; and what the evidence supports about an individual's abilities. Agreement on one does not settle the others. Score interpretations and reference populations need explicit support (American Educational Research Association et al., 2014, Standards 1.1–1.4 and 5.8–5.9).

An eligibility probability refers to the designated population before application or membership selection. Correlations measured among applicants or members need not equal the population correlation used in equation (1). Selection can change the relationship, and adjustments depend on the selection mechanism (Sackett and Yang, 2000).

A procedure accepting the best result from repeated attempts also differs from a fixed two-test procedure. For m pass events, independently of any Gaussian model,

$$\max_i P(A_i) \le P\left(\bigcup_{i=1}^{m} A_i\right) \le \min\left(1, \sum_{i=1}^{m} P(A_i)\right). \tag{15}$$

If attempts were independent with identical pass probability p, the union would be $1 - (1 - p)^m$. Actual retests usually require a model of dependence and administration history; practice and coaching can also change score distributions (Hausknecht et al., 2007). An analysis should calibrate the procedure actually used, including its attempts, stopping rule, and accepted instruments.

Towers was right to ask how multiple admission routes affect eligibility, and right to withdraw a faulty calculation. A defensible replacement states the rule, the reference population, the joint-distribution model, the thresholds, and each conditional denominator. Its conclusions remain conditional on those specifications.

A Numerical audit of the follow-up

The BASIC listing uses $\rho = 0.7$, means 100, standard deviations 16, and grid nodes from Q through 199 in steps of $h = 0.5$, with the full-field denominator starting at 1. In mathematical notation its probability estimate is

$$\hat{J}_Q = \frac{\sum_{x\in\{Q,Q+h,\ldots,199\}} \sum_{y\in\{Q,Q+h,\ldots,199\}} g(x,y)}{\sum_{x\in\{1,1+h,\ldots,199\}} \sum_{y\in\{1,1+h,\ldots,199\}} g(x,y)}, \tag{16}$$

where g is proportional to the Gaussian density on the IQ-point coordinates. The full grid has $397^2 = 157{,}609$ nodes. A constant factor $1/(16 \cdot 16)$ and a cell area h^2 would cancel between the two sums. The corresponding absent scale factor also cancels in the displayed double-integral ratio in the follow-up.

Table 3: Reproduction of the follow-up's grid estimates. Entries are probabilities, not percentages. The last column integrates from the actual continuous cutoff Q, on both tests.

Q	**Published grid ratio**	**Reproduced grid ratio**	**Continuous joint probability**
133	0.0064061692	0.0064061681	0.0061136512
150	0.00014748463	0.00014748460	0.00013830591
164	2.6393845×10^{-6}	2.6393840×10^{-6}	2.4390159×10^{-6}

The reproduced and published grid values agree to relative differences of about 2×10^{-7}, but not to every printed digit. The exact cause of those small differences is not established. They are far smaller than the difference between the grid estimate and the continuous probability at the intended cutoff.

Equal weighting at the boundary includes a full contribution from the node at Q. Viewed as midpoint quadrature, its first cell begins at $Q - h/2 = Q - 0.25$. The effective lower boundary is therefore shifted downward. The finite upper boundary and ordinary quadrature error add smaller discrepancies in these examples. If IQ scores were explicitly rounded to integers before classification, a separate rounding rule would be needed; a quarter-point grid shift cannot simply be assumed to be that rule.

For continuous scores, the actual marginal exceedance probabilities at the stated integer cutoffs are those in table 4. Dividing the continuous joint tail by its matching marginal gives a consistent conditional probability.

Table 4: Consistent continuous calculations at the follow-up's integer SD-16 cutoffs, with $\rho = 0.7$. All probability columns are percentages.

Cutoff	**Marginal pass rate**	**Both: population**	**Second pass given first**
133	1.95800788	0.61136512	31.223834
150	0.08890253	0.01383059	15.557028
164	0.00316712	0.00024390	7.701043

The follow-up divides its grid values by 0.02, 0.001, and 0.000030303. Those are approximations to desired nominal tails, not matching marginal tails of the specific grid or continuous integer cutoffs. Its approximately 32%, 15%, and 9% summaries consequently mix conventions. For exact nominal percentiles, use

table 2; for actual continuous integer cutoffs, use table 4. Using the grid's own marginal denominator gives approximately 31.51%, 15.74%, and 7.81%, a third, internally defined approximation.

B Reproducibility and numerical checks

The accompanying script evaluates equation (3) with adaptive one-dimensional quadrature and directly uses inverse survival functions for the thresholds. SciPy documents both its normal survival functions and its integration routine (SciPy contributors, n.d.a,n).

An independent integration identity supplies a cross-check:

$$J(a,b;\rho) = \overline{\Phi}(a)\overline{\Phi}(b) + \int_0^\rho \frac{\exp\left[-\dfrac{a^2 - 2tab + b^2}{2(1-t^2)}\right]}{2\pi\sqrt{1-t^2}}\,dt. \tag{17}$$

This follows by integrating the Gaussian probability's derivative with respect to its correlation, starting at independence. Checks also cover symmetry, probability bounds, the original mixed-case union violation, the binary phi coefficient, the calibrated OR threshold, and reproduction of the follow-up's grid.

The CSV files contain unrounded model calculations to complement the rounded display tables. Run `make check` from this analysis folder; `make pdf` builds the report with XeLaTeX. Numerical consistency does not establish the empirical truth of the Gaussian assumptions or the chosen correlation.

C Expanded verification results and limits

The original proof selected only `Admission_Core`. The expanded implementation also puts `Normal_Model`, `Towers_Engine`, and the `Towers_Demo` main procedure in SPARK, apart from small explicit numerical completions. Parsing and CSV/file output are isolated in `Towers_IO`, with external state modeled in its SPARK interface. The proof project now enumerates all nine source files, including the trusted I/O completion. Turning an aspect on is a language-coverage change; the fresh native proof results below provide the separate verification evidence.

The expanded computer10 campaign was completed on 1 November 2026. Its directory retains the date on which the expansion began. It records exact commands, tool versions, source hashes, native reports and raw coverage, checksums, numerical comparisons, and Python results. Earlier 31-check and 110-test results remain immutable historical snapshots; they are not totals for this expanded implementation.

C.1 SPARK scope and results

The fresh five-prover portfolio discharged **434 checks: 91 flow checks and 343 prover checks**, with zero unproved and zero justified checks. CVC5, Z3, Alt-Ergo, classic COLIBRI, and Yices were requested together. A completed portfolio does not imply that every solver completed every independent obligation. The deliberately incorrect union control was rejected with an unproved functional contract. All invocations use fresh source copies and object directories. The matching toolchain reports GNAT 13.2.0 and GNATprove banner `0.0w`; its executable hashes are recorded in the manifest.

Native coverage records 36 `all`, 16 `spec`, and zero `no` entities. The statuses describe analyzed bodies and trusted specifications, not successful proofs by themselves. Whole off-mode units can emit empty native entity arrays, so zero `no` entries do not mean that no code is excluded. The explicit earlier-source baseline

has 11 `all` entities; it records flow/coverage only and cannot count as a new proof. The auditor separately checks native proof totals, complete metadata, and coverage of each required driver and numerical body.

The original core contracts still establish classification into the four event cells, exactly one increment per observation, bounded marginal and union counts, and exact integer/rational inclusion-exclusion, complements, and conditional fractions. Expanded SPARK code covers inverse-tail bisection, Simpson summation and grid loops, the exact Park–Miller state transition and uniform conversion, Box–Muller composition, calculations, simulation/count loops, Wilson arithmetic, result assembly, and the main procedure. Joint-tail contracts require $0 \le \rho \le 0.95$, positive even panel counts, and $|A|, |B| \le 10^{100}$; inversion requires a probability strictly inside the endpoint survival bracket. The main cases satisfy these implementation domains. Small SPARK helpers and proved loop invariants bound accumulation without importing proof axioms or changing the numerical summation order.

C.2 Checked interfaces and numerical limitations

Small `Normal_Model` completions remain off for square root, normal survival via foreign `erfc`, logarithm and exponential ranges, exponent and quadrature node validation, probability bounds, and grid normalization. The engine's checked completions validate quantization error and the joint, union, and conditional relationships needed by the exact core. Each promised normal-return bound or relationship is checked before returning; an invalid value raises an error rather than being clamped or changed. `Towers_IO`'s parsing, formatting, filesystem operations, and exceptions remain outside the proof.

Analysis uses these checked interfaces and the installed Ada elementary-function specifications, whose contracts support SPARK but provide limited numerical information. A successful caller proof does not prove the implementations, that a numerical guard can never reject, or that I/O succeeds. There are no project `pragma Assume` statements, imported proof axioms, or proof suppressions. Zero explicit assumptions or suppressions in native metadata does not remove trust in off-mode completions and standard-library contracts. No proof of inversion accuracy, survival-function monotonicity, quadrature error, or a Gaussian distribution is claimed. The 10,000/20,000-panel comparison remains a numerical refinement check, not a certified integration-error bound.

The floating-to-core conversion rounds probabilities to units of 10^{-15} and checks reconstruction error against $\delta = 0.51 \times 10^{-15}$. For a conditional probability j/p, the resulting ratio change is bounded by $\delta(1 + j/p)/(p - \delta)$, for $p > \delta$. At the smallest nominal marginal $p = 1/33{,}000$, this is at most approximately 3.37×10^{-11}; the checked changes are smaller. This accounts only for quantization. Raw floating union and conditional values are retained alongside the exact-core results.

C.3 Numerical regression, Deal, and evidence audits

A fresh Ada build reproduces all four retained CSVs byte for byte. Independent SciPy validation and Python replay pass normally and under `python -O`. One million shared correlated pairs from seed `20261030` supply the four cases; all 16 event-cell counts match exactly. Minimum valid trial and boundary-seed runs also succeed, and invalid options are rejected. Replay checks reproducibility and agreement, not random independence. Wilson intervals describe simulation uncertainty under the model; rare events at this sample size cannot establish extreme-tail psychometric frequencies.

All current Python files and named functions, including tests, fixtures, and evidence tools, carry Deal contracts. The final repository-wide inventory has 22 files, 177 named functions, and 502 Deal decorators with zero gaps. All 21 positive files have zero lint diagnostics, and 164 tests pass in each execution mode. Coverage, clean positive static lint, the exact expected `DEL012` negative control, runtime/property tests, numerical scripts, and historical-source audits pass in normal and optimized Python. CLI entry points

explicitly enable Deal under optimization. Deal Solver proves the two pure proof-acceptance gate post-conditions and rejects the incorrect symbolic control. These focused results do not prove the full Python programs, SciPy, or subprocess handling; clean lint and finite tests are not universal proofs.

From a full Git clone, install the pinned dependencies and run `make check`; run `verification/verify_evidence.py` to audit the current expanded campaign. The auditor requires both production projects, complete native coverage metadata, required SPARK-on bodies, consistent native totals, source identities, checksums, and rejection of the negative control. Historical Ada campaigns require an explicit matching old source tree; changed Ada cannot silently fall back to history. Historical Python identities may be resolved through verified Git blobs. Audits check record integrity and identity; they do not rerun the provers or independently validate SMT proof objects.

None of these software checks establishes that $\rho = 0.7$ applies to actual tests, that empirical norms and tails are accurate, or that model eligibility probabilities describe real membership selection or latent ability. Those empirical and modeling questions remain as identified in the main analysis.

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